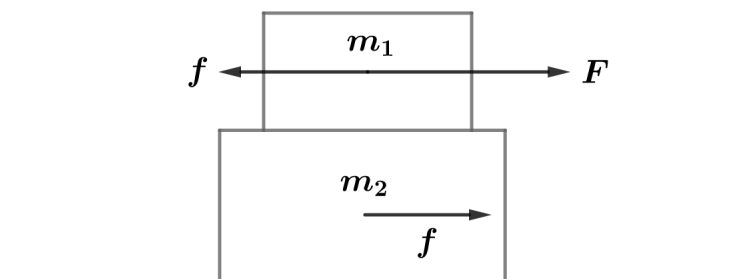


2018B F=ma Exam: Problem 13

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Applying $F=ma$ to the entire system, we have

$$F = (m_1 + m_2)a$$

Applying $F=ma$ to m_1 , we have

$$F - f = m_1a$$

When m_1 is about to slip, $f = \mu N = \mu m_1 g$:

$$F - \mu m_1 g = m_1 \left(\frac{F}{m_1 + m_2} \right)$$

$$\left(\frac{m_2}{m_1 + m_2} \right) F = \mu m_1 g$$

Hence,

$$F = \left(\frac{m_1 + m_2}{m_2} \right) \mu m_1 g = 12.0 \text{ N}$$

so the answer is D.